



SCUD ARSENAL

MAGAZINE



Cybersecurity



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Dear Readers,

Greetings and warm welcome to June 2018 issue Volume9 Issue1 of our Magazine , "SCUD ARSENAL".

The previous semester witnessed various ecstatic moments as Autonomous status is given to our college and our department got NBA Accreditation in the month of march. We are looking forward to commence the forthcoming semester with ecstasy and earnest commitment.

Best Wishes
Editorial Board

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SWIFT

Development of Swift started in July 2010 by Chris Lattner, with the eventual collaboration of many other programmers at Apple. Swift took language ideas "from Objective-C, Rust, Haskell, Ruby, Python, C#, CLU, and far too many others to list". On June 2, 2014, the Apple Worldwide Developers Conference (WWDC) application became the first publicly released app written with Swift.[22] A beta version of the programming language was released to registered Apple developers at the conference, but the company did not promise that the final version of Swift would be source code compatible with the test version. Apple planned to make source code converters available if needed for the full release.

The Swift Programming Language, a free 500-page manual, was also released at WWDC, and is available on the iBooks Store and the official website.

Swift reached the 1.0 milestone on September 9, 2014, with the Gold Master of Xcode 6.0 for iOS. Swift 1.1 was released on October 22, 2014, alongside the launch of Xcode 6.1. Swift 1.2 was released on April 8, 2015, along with Xcode 6.3. Swift 2.0 was announced at WWDC 2015, and was made available for publishing apps in the App Store in September 21, 2015. Swift 3.0 was released on September 13, 2016. Swift 4.0 was released on September 19, 2017. Swift 4.1 was released on March 29, 2018.

Swift won first place for Most Loved Programming Language in the Stack Overflow Developer Survey 2015 and second place in 2016.

In December 2015, IBM announced its Swift Sandbox website, which allows developers to write Swift code in one pane and display output in another.

During the WWDC 2016, Apple announced an iPad exclusive app, named Swift Playgrounds, intended to teach people how to code in Swift. The app is presented in a 3D video game-like interface which provides feedback when lines of code are placed in a certain order and executed.

In January 2017, Chris Lattner announced his departure from Apple for a new position with Tesla Motors, with the Swift project lead role going to team veteran Ted Kremenek.

Swift is a fantastic way to write software, whether it's for phones, desktops, servers, or anything else that runs code. It's a safe, fast, and interactive programming language that combines the best in modern language thinking with wisdom from the wider Apple

engineering culture and the diverse contributions from its open-source community. The compiler is optimized for performance and the language is optimized for development, without compromising on either.

Swift is friendly to new programmers. It's an industrial-quality programming language that's as expressive and enjoyable as a scripting language. Writing Swift code in a playground lets you experiment with code and see the results immediately, without the overhead of building and running an app.

Swift defines away large classes of common programming errors by adopting modern programming patterns:

- Variables are always initialized before use.
- Array indices are checked for out-of-bounds errors.
- Integers are checked for overflow.
- Optionals ensure that nil values are handled explicitly.
- Memory is managed automatically.

Error handling allows controlled recovery from unexpected failures.

Swift code is compiled and optimized to get the most out of modern hardware. The syntax and standard library have been designed based on the guiding principle that the obvious way to write your code should also perform the best. Its combination of safety and speed make Swift an excellent choice for everything from "Hello, world!" to an entire operating system.

Swift combines powerful type inference and pattern matching with a modern, lightweight syntax, allowing complex ideas to be expressed in a clear and concise manner. As a result, code is not just easier to write, but easier to read and maintain as well.

Swift has been years in the making, and it continues to evolve with new features and capabilities. Our goals for Swift are ambitious. We can't wait to see what you create with it.

- Article By
M.C.J Swaroop(16A81A0599)

Cockroach DB

HISTORY:-

George Beccaloni began work on the database that was to become the CSF in 2004. This early version simply contained a list of all currently valid cockroach genera and their higher classification. In February 2005 Daniel Otte (USA) suggested to George that he might consider developing a comprehensive Species File database for cockroaches and he put George in contact with David Eades (USA) so that they could discuss this possibility.

George finished work on the first version of the CSF database in January 2007 and he sent it to David and his team for migration from MS Access to the Species File system. Most of George's work on this database was done in his spare time during 2006 (George's current job does not allow him official time to work on this project). The CSF first went live on the Web in May 2007 and data migration was completed in July 2007.

Since then George has spent as much of his 'spare' time as he can afford correcting and editing data in the CSF and adding information from recently published papers. In 2007 George obtained funding from the Orthopterists' Society to photograph types and other specimens in the London Natural History Museum's (NHM) collection. Ed Baker was employed to do this work and he imaged a total of 219 types (153 primary and 66 secondary types) plus 283 non-types in the time available. In 2011 funding was obtained from the 4D4Life project via Chris Lyal (NHM) to employ Paul Brock to check c. 500 species-group and infrasubspecific names which appeared to be absent from the CSF. The names were carefully checked and the few which were actually missing were entered. Paul also uploaded a large number of high quality images of NHM specimens taken by Zongqing Wang (China), who spent one year working with me on the cockroach collection at the NHM.

Note that from 2007 to June 2014 the CSF was named the Blattodea Species File. The name was changed to Cockroach Species File in July 2014.

Quality of CSF data

The CSF contains all published genus- and species-group names and infrasubspecific names, except for a small number published in recent papers which have not been entered yet. Very few older names are missing and these will be added when they are discovered.

So far, most of the effort put into error checking data in the CSF has concentrated on ensuring that names of taxa are correctly spelt and that the author names and years of publication are correct. Numerous errors in Princis' catalogue have been corrected and a number of names have been added which were discovered to be missing from his catalogue.

Unfortunately Princis' catalogue does not always give the original published spellings and combinations of names and sometimes when this information is present it is ambiguous or incorrect. Correcting this data is an on-going task and it will ultimately be necessary to examine the original descriptions of all taxa in order to ensure that this information is accurately recorded in the CSF.

An important point to note is that the references given to the publications in which names were originally published are incorrect for a small proportion of the taxa in the CSF. This problem resulted from having to link a database containing the names of taxa to a database containing taxonomic

references. Mismatches occurred in some cases where the author of a name published more than one taxonomic paper in the same year as the name in question. These errors are being corrected when they are discovered.

Future development

Work on the CSF will be prioritised as follows:

Links between names of taxa and references to original descriptions will be checked and corrected.

Full citations will be entered for currently incomplete references to original descriptions.

All original descriptions will be checked to ensure that the page number, original spelling of the name and the original combination of the name are correctly recorded in the CSF. Information about type specimens (kind of type(s), numbers of types, their sexes, the localities they were collected from, and their depositories) plus information about species/subspecies distributions will be entered into the CSF.

Synonyms and references will be added for family-group names.

Pdf files of out of copyright references will be made available for downloading.

Article By-

Y.Yuva Sri Durga(16A81A05D7)

LARAVEL

Laravel is a free, open-source[3] PHP web framework, created by Taylor Otwell and intended for the development of web applications following the model–view–controller (MVC) architectural pattern and based on Symfony. Some of the features of Laravel are a modular packaging system with a dedicated dependency manager, different ways for accessing relational databases, utilities that aid in application deployment and maintenance, and its orientation toward syntactic sugar.

The source code of Laravel is hosted on GitHub and licensed under the terms of MIT License.

Taylor Otwell created Laravel as an attempt to provide a more advanced alternative to the CodeIgniter framework, which did not provide certain features such as built-in support for user authentication and authorization. Laravel's first beta release was made available on June 9, 2011, followed by the Laravel 1 release later in the same month. Laravel 1 included built-in support for authentication, localisation, models, views, sessions, routing and other mechanisms, but lacked support for controllers that prevented it from being a true MVC framework.

Laravel 2 was released in September 2011, bringing various improvements from the author and community. Major new features included the support for controllers, which made Laravel 2 a fully MVC-compliant framework, built-in support for the inversion of control (IoC) principle, and a templating system called Blade. As a downside, support for third-party packages was removed in Laravel 2.

Laravel 3 was released in February 2012 with a set of new features including the command-line interface (CLI) named Artisan, built-in support for more database management systems, database migrations as a form of version control for database layouts, support for handling events, and a packaging system called Bundles. An increase of Laravel's userbase and popularity lined up with the release of Laravel 3.

Laravel 4, codenamed Illuminate, was released in May 2013. It was made as a complete rewrite of the Laravel framework, migrating its layout into a set of separate packages distributed through Composer, which serves as an application-level package manager. Such a layout improved the extensibility of Laravel 4, which was paired with its official regular release schedule spanning six months between minor point releases. Other new features in the Laravel 4 release include database seeding for the initial population of databases, support for message queues, built-in support for sending different types of email, and support for delayed deletion of database records called soft deletion.

Laravel 5 was released in February 2015 as a result of internal changes that ended up in renumbering the then-future Laravel 4.3 release. New features in the Laravel 5 release include support for scheduling periodically executed tasks through a package called Scheduler, an abstraction layer called Flysystem that allows remote storage to be used in

the same way as local file systems, improved handling of package assets through Elixir, and simplified externally handled authentication through the optional Socialite package. Laravel 5 also introduced a new internal directory tree structure for developed applications.

Article By

S.V Subbarao(15A81A0582)

CHATBOT

A chatbot (also known as a talkbot, chatterbot, Bot, IM bot, interactive agent, or Artificial Conversational Entity) is a computer program or an artificial intelligence which conducts a conversation via auditory or textual methods. Such programs are often designed to convincingly simulate how a human would behave as a conversational partner, thereby passing the Turing test. Chatbots are typically used in dialog systems for various practical purposes including customer service or information acquisition. Some chatterbots use sophisticated natural language processing systems, but many simpler systems scan for keywords within the input, then pull a reply with the most matching keywords, or the most similar wording pattern, from a database.

The term "ChatterBot" was originally coined by Michael Mauldin (creator of the first Verbot, Julia) in 1994 to describe these conversational programs. Today, most chatbots are either accessed via virtual assistants such as Google Assistant and Amazon Alexa, via messaging apps such as Facebook Messenger or WeChat, or via individual organizations' apps and websites. Chatbots can be classified into usage categories such as conversational commerce (e-commerce via chat), analytics, communication, customer support, design, developer tools, education, entertainment, finance, food, games, health, HR, marketing, news, personal, productivity, shopping, social, sports, travel and utilities.

HOW CHATBOTS WORK

Chatbots such as ELIZA and PARRY were early attempts at creating programs that could at least temporarily fool a real human being into thinking they were having a conversation with another person. PARRY's effectiveness was benchmarked in the early 1970s using a version of a Turing test: testers only made the correct identification of human vs. chatbot at a level consistent with making a random guess.

Chatbots have come a long way since then. They are built on AI technologies, including deep learning, natural language processing and machine learning algorithms, and require massive amounts of data. The more an end user interacts with the bot, the better voice recognition becomes at predicting what the appropriate response is when communicating with an end user.

Chatbots can be stateless or stateful. A stateless chatbot approaches each interaction as if it was with a new user. A stateful chatbot is more sophisticated; it can review past interactions and frame new responses in context. Adding a chatbot to a company's service or sales department today requires little coding, because there are a number of chatbot service providers that allow developers to build conversational interfaces for any business application.



APPLICATIONS

➤ MESSAGING APPS

Many companies' chatbots run on messaging apps like Facebook Messenger (since 2016), WeChat (since 2013), WhatsApp, LiveChat, Kik, Slack, Line, Telegram, or simply via SMS. They are used for B2C customer service, sales and marketing. In 2016, Facebook Messenger allowed developers to place chatbots on their platform. There were 30,000 bots created for Messenger in the first six months, rising to 100,000 by September 2017.

Since September 2017, this has also been as part of a pilot program on WhatsApp. Airlines KLM and Aeroméxico both announced their participation in the testing both airlines had previously launched customer services on the Facebook Messenger platform. The bots usually appear as one of the user's contacts, but can sometimes act as participants in a group chat.

Many banks and insurers, media and e-commerce companies, airlines and hotel chains, retailers, health care providers, government entities and restaurant chains have used chatbots to answer simple questions, increase customer engagement,[23] for promotion, and to offer additional ways to order from them.

➤ AS PART OF COMPANY APPS AND WEBSITES

Previous generations of chatbots were present on company websites, e.g. Ask Jenn from Alaska Airlines which debuted in 2008 or Expedia's virtual customer service agent which launched in 2011. The newer generation of chatbots includes IBM Watson-powered "Rocky", introduced in February 2017 by the New York City-based e-commerce company Rare Carat to provide information to prospective diamond buyers.

➤ COMPANY INTERNAL PLATFORMS

Other companies explore ways they can use chatbots internally, for example for Customer Support, Human Resources, or even in Internet-of-Things (IoT) projects. Overstock.com, for one, has reportedly launched a chatbot named Mila to automate certain simple yet time-consuming processes when requesting for a sick leave. Other large companies such as Lloyds Banking Group, Royal Bank of Scotland, Renault and Citroën are now using automated online assistants instead of call centres with humans to provide a first point of contact.

➤ TOYS

Chatbots have also been incorporated into devices not primarily meant for computing such as toys. Hello Barbie is an Internet-connected version of the doll that uses a chatbot provided by the company ToyTalk, which previously used the chatbot for a range of smartphone-based characters for children, storyline.



Article By

S.Karthik(15A81A05G8)

Fun Corner

Questions:-

1. If a boy is slapped by his mother & his father, which one hurts the most?
2. There were school of 10 fish in an enclosed tank. 3 of them sank 4 swam away and 3 died. How many fish are there now?
3. If a man were born in Greece, Raised in Spain, came to America, and died in San Francisco. What is he?
4. What is the first thing common man do in the morning?
5. The pilot said that Robert was their brother. Robert said he had no brother who is lying?
6. Which side of a cat has the most fur?
7. All 5 boys are busy in a room. Sam is reading a book, Alex is cooking, Max is playing chess, John is doing the laundry. What is the fifth boy doing?
8. The more you take, the more you leave behind what are they?
9. What never asks questions but is often answered?
10. Name four days of the week that start with the letter 'T'?
11. What can run but can't walk?
12. What has hands but can not clap?
13. In North Pole, a house has 3 walls. All of the walls are facing south, and a bear is circling the house. What color is the bear?
14. A man is sitting in a house at night that has no lights on at all. There is no lamp, no candle, nothing, yet he is reading. How?
15. Imagine you're in a room that is filling with water. There are no windows or door, you cannot break anything on the room. How do you get out?

Sudoku:-

	2		5		1		9	
8			2		3			6
	3			6			7	
		1				6		
5	4						1	9
		2				7		
	9			3			8	
2			8		4			7
	1		9		7		6	

Details of faculty attended FDPs, Workshops and Seminars

S.No	Name of the Faculty	Name of Workshop/Seminar/ FDP/SDP	Location	No. Of Days	From Date	To Date
1.	Ch.Raja Ramesh	Artificial intelligence and deep learning	S.R.K.R Engineering college Bhimavarm	03	28-06-2018	30-06-2018
2.	K.Shirin Bhanu	Artificial intelligence and deep learning	S.R.K.R Engineering college Bhimavarm	03	28-06-2018	30-06-2018
3.	G.Loshma	Artificial intelligence and deep learning	S.R.K.R Engineering college Bhimavarm	03	28-06-2018	30-06-2018
4.	D.Ganesh	Artificial intelligence and deep learning	S.R.K.R Engineering college Bhimavarm	03	28-06-2018	30-06-2018
5.	K.V.Muthyalu	Artificial intelligence and deep learning	S.R.K.R Engineering college Bhimavarm	03	28-06-2018	30-06-2018
6.	B.Sri Ramya	FDP on Pedagogical Methodologies	Sri Vasavi Engineering CollegePedatadepalli	06	18-06-2018	23-06-2018
7.	M.Nageswara Rao	FDP on Pedagogical Methodologies	Sri Vasavi Engineering CollegePedatadepalli	06	18-06-2018	23-06-2018
8.	A Rajesh	Statistics with R programming	Pragathi Engineering college,Surampalem	06	18-06-2018	23-06-2018

S.No	Name of the Faculty	Name of Workshop/Seminar/ FDP/SDP	Location	No. Of Days	From Date	To Date
9.	G Sri ram Ganesh	Internet of Things by Electronic and ICT Academies	NIT Warangal	05	18-06-2018	22-06-2018
10.	K Lakshmi Narayana	Internet of Things by Electronic and ICT Academies	NIT Warangal	05	18-06-2018	22-06-2018
11.	T Sujith Kumar	ANN and Deep Learning by Electronic and ICT Academies	NIT Warangal	05	11-06-2018	15-06-2018
12.	Syed sabir Hussain	AI and Machine Learning by Electronic and ICT Academies	NIT Warangal	05	04-06-2018	08-06-2018
13.	N B Praveen Kumar	AI and Machine Learning by Electronic and ICT Academies	NIT Warangal	05	04-06-2018	08-06-2018

Details of faculty attended Conferences

S.No	Name and Designation of the Faculty	Name of the conference	Location	No. Of Days	From Date	To Date
1	Dr. D. Jaya Kumari	Transformations in Engg Education-Imparting Futuristic Skills	SRM University, Amaravathi	03	15-07-2018	17-07-2018

Workshop conducted by the Department

S.No	Date	Event Name	Name of the Eminent Guest	Audience	No of students participated
1	09-07-2018 to 14-07-2018	Google Android Developer Fundamentals Workshop by APSSDC	1. Ms.P.Prabhu Sandhya 2. Mr.U.Venkatesh 3. Ms.N.Devi Anusha	III Yr. CSE-A Students	63

Details of Visitors who visited the Department

S.No	Name of Eminent Guest	Organization	Date(s) of Visit
1.	Dr.R B V Subramanyam	Professor, NITW	02.06.2018
2.	Sri. Vuppalapati Srinivasa Raju	Director, Alykas Software Solutions	02.06.2018
3.	Mr.Eedala Rambabu	Micofocus, Bangalore	02.06.2018

Guest Lectures Conducted for Students

S.No	Name of the Event	Speaker institute	Date	Audience	Organized By
1	Engaging New age Talent-Wants, Needs and Values	Mr.Ganesh Somisetty, Air Worldwide, HR Manager	28.07.2018	IV-A,B,C &D	Department
2	Personality Development Skills	Dr.G Rajashekhar, i-focus	18.07.2018	IV-A,B,C &D	College
3	Indian Innovations	Er.T. Praveen Kumar, Senior Embedded System Engineer, Robert BOSCH, Bangalore	13-07-2018	III year(Selected Students)	College
4	Benefits of becoming a member of IE (India) "	Er. Sudeep Chakravarthy , Assistant Director , Technical, Institution of Engineers India IE (I)	03-07-2018	II year (Selected Students)	College

Guest Lectures conducted for Faculty

S.No	Name of the Event	Speaker institute	Date	Audience	Organized By
1	Measures to strengthen the Quality of Academics	Prof. S.R.K. Reddy , Professor of CE and Advisor to the management, Gudlavalluru Engineering College	07-07-2018	Faculty (46)	College

Details of faculty publications

S.No	Name of the Faculty	Publication Title	Publication Details
1.	Mrs Loshma Guniseti	An Efficient Approach for Multi-Dimensional Dataset Search using Keywords	International Journal of Scientific Development and Research (IJS DR)-June 2018 Volume 3, Issue 6 ISSN: 2455-2631
2.	Dr Vedula Venkateswara Rao	Performance of Memory Virtualization Using Local Memory Resources Balancing	International journal of Engineering Research and Technology
3.	Dr Vedula Venkateswara Rao	Mining Frequent Patterns From Big Data sets using Genetic Algorithm	International Journal of Engineering Research and Technology , Volume 11 No2
4.	Dr Vankamamidi. S. Naresh	Provable Secure Lightweight Hyper Elliptic Curve Based Communication System for Wireless Sensor Networks	International Journal of Communication Systems

Faculty Achievements

Our department faculty and Associate Dean, R&D, Dr. V.S.Naresh, published an Article entitled: "Provable Secure Lightweight Hyper Elliptic Curve Based Communication System for Wireless Sensor Networks" published in International Journal of Communication Systems (FREE SCI JOURNAL with impact factor of 1.717 and Publisher: John Wiley & Sons Ltd and Online ISSN:1099-1131) in the name of our Sri Vasvi Engineering College on 29 July 2018

First Year Orientation Programme

B.Tech First year orientation programme has been conducted by our department from 9-07-2018 to 21-07-2018 for 2018-19 Admitted batch. Our department senior faculty and third year and final year students presented exquisite seminars to inspire first year students.

Collaboration with LEADINGINDIA.AI

Our College got collaborated with LEADINGINDIA.AI, Bennett University, Greater Noida which has taken a Nation-wide initiative on AI and deep learning Skilling and Research and Our college is the Zonal Lead Partner for the period of June 2018 to March 2020

This project will Skill 10000 faculty/professionals of 1000 institutions in India through 10 collaborators 100 Zonal Lead institutions chosen from different Geographic regions of the country. 100 Zonal Lead institutions will also setup a research group, which will be mentored to do short and long-term research in Machine Learning, AI and Deep learning. Start-ups and industry involved in the Artificial Intelligence related work will partner for providing application-inspired ideas and absorbing the trained graduates of these institutions.

Our Department members created a Research Group with the name ATOM (Artificial Intelligence TO Machine Learning)

Details of Research Group Members

S.No.	Faculty Name	Designation	Role
1	Dr. K. Shirin Bhanu	Associate Professor	Team Lead
2	G. Loshma	Associate Professor	Member
3	Ch.Raja Ramesh	Associate Professor	Member
4	K.Venkata Mutyalu	Assistant Professor	Member
5	D.Ganesh	Assistant Professor	Member
6	Dr.J. Veera Raghavan	Associate Professor	Additional Contributor

The Student Ambassador of AI Club is GSK Chaitanya of III B.Tech CSE

Other details:

- ❖ **Research Domain:** Agriculture
- ❖ **Project:**
 - Automated quality assessment of crops
 - Crop type classification using image classification
- ❖ **Targets for Research group by the end of two years:**
- ❖ **Research:**
 - No of Publications : 2
 - Funding Grants to be applied : 2
 - M.Tech Student projects related to AI: 1

Development:

- New datasets creation for deep learning applications : 2
- Products/Applications(students, Research Scholars): 1
- Resources to be produced : 2
- Industries Experts to be contacted for our Product: 2
- **Activities done:**
 - Attended 3 day **AI and Deep Learning** Workshop at SRKR Engineering College during 28-30th June 2018.
 - External Review-1 completed on 28th June 2018.
 - Conducted **AI and Deep Learning Orientation** for the first year B.Tech CSE students during 9-21st July 2018.

Student Achievements:-

Below are just a few of the activities:

- a) Workshops & Challenges
- b) Campus Engagement
- c) Spaces
- d) Student Organizations & Communities
- e) Community Engagement
- f) Hackathons & Makeathons
- g) Campus Resources
- h) First-Year Engagement
- i) Classes
- j) Talks & Speaker Series
- k) Global Leadership Opportunities

SAHAYA

Our Sahaya team donated various food materials, medicines and clothes with an amount of 10000 rupees to kerala floods affected people



ARTIFICIAL INTELLIGENCE /DEEP LEARNING

Artificial Intelligence

Artificial Intelligence is an area of computer science; it has the capability of machine to reproduce intellectual human behavior. Artificial Intelligence is almost mimicking the human brain, by understanding how it thinks, how it behaves, how it learns, take decisions and work while problem solving. AI is mainly depends on data set which is created with existing experiences.

For example: A machine playing chess, or a voice activated software which helps you with various things in your iPhone or a Number plate Recognition system which captures the number plate of a over speeding car and processes it to extract the registration number and identify the owner of the car.



Machine Learning is subset of Artificial intelligence and deep learning is subset of Machine Learning which is shown in the above Fig. 1.

Machine Learning

Machine Learning is a subset of Artificial Intelligence which provides computers with the ability to learn without being explicitly programmed. In machine learning, we do not have to define explicitly all the steps or conditions like any other programming application. On the contrary, the machine gets trained on a training dataset, large enough to create a model, which helps machine to take decisions based on its learning.

Examples of Machine Learning

One of the best examples is Face book's News feed. News feed uses machine learning to personalize each members feed. If a member frequently stops scrolling to read or like a particular friend's posts, the News Feed will start to show more of that friend's activity earlier in the feed. Behind the scenes, the software is simply using statistical analysis and predictive analytics to identify patterns in the user's data and use those patterns to populate the News Feed. Should the member no longer stop to read, like or comment on the friend's posts, that new data will be included in the data set and the News Feed will adjust accordingly?

The following Fig. 2 represents the determine the species of a flower based on its petal and sepal length (leaves of a flower using Machine Learning)

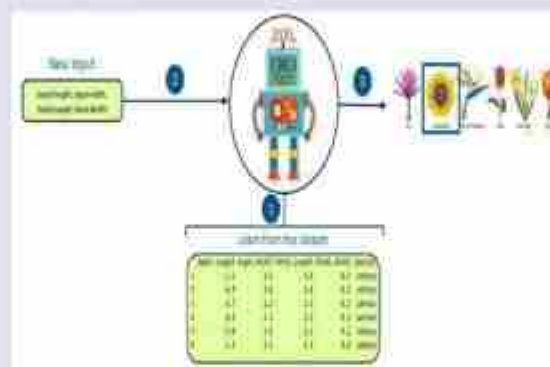


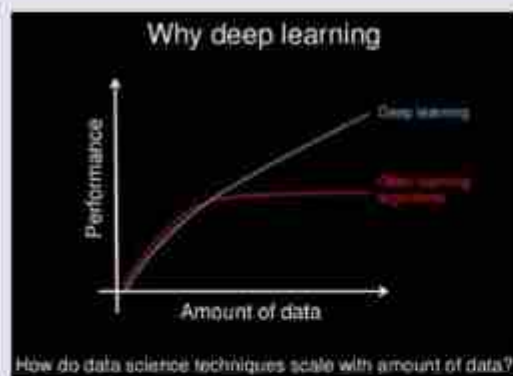
Fig 2.

Limitations of Machine Learning

Machine Learning is not capable of handling high dimensional data that is where input & output is quite large. Handling and processing such type of data becomes very complex and resource exhaustive. This is termed as Curse of Dimensionality.

Deep Learning

Practically, Deep Learning is a subset of Machine Learning that achieves great power and flexibility by learning to represent the world as nested hierarchy of concepts, with each concept defined in relation to simpler concepts, and more abstract representations computed in terms of less abstract ones [2]. For most flavors of the old generations of learning algorithms performance will plateau deep learning is the first class of algorithms that is scalable performance just keeps getting better as you feed them more data. It is shown in the figure 3[1].



How Deep Learning Works?

In an attempt to re-engineer a human brain, Deep Learning studies the basic unit of a brain called a brain cell or a neuron. Inspired from a neuron an artificial neuron or a Perceptron was developed. Now, let us understand the functionality of biological neurons and how we mimic this functionality in the Perceptron or an artificial neuron.

While deep learning was first theorized in the 1980s, there are two main reasons it has only recently become useful:

- Deep learning requires large amounts of labeled data. For example, driverless car development requires millions of images and thousands of hours of video.
- Deep learning requires substantial computing power. High-performance GPUs have a parallel architecture that is efficient for deep learning. When combined with clusters or cloud computing, this enables development teams to reduce training time for a deep learning network from weeks to hours or less.

Applications of Deep Learning:

We have So many Inspirational Applications of Deep Learning in that some of the applications are listed below[3][4]

- Self driving cars.
- Deep learning in Health care
- Voice search and voice activated assistants.
- Automatic text writing
- Colorization of Black and White Images.
- Adding Sounds To Silent Movies.
- Automatic Machine Translation.
- Object Classification in Photographs.
- Automatic Handwriting Generation.
- Character Text Generation.
- Image Caption Generation.
- Automatic Game Playing.

Tools Required for Deep Learning:

So many open source tools are available to implement deep learning; some of the tools are listed below.

- Keras
- TensorFlow using Google Colab
- SciKit-Learn
- Pandas
- Nltk
- Numpy
- Edward
- Lime
- Endnotes
- References:

1.<https://machinelearningmastery.com/what-is-deep-learning/>

2.<https://towardsdatascience.com/why-deep-learning-is-needed-over-traditional-machine-learning-1b6a99177063>.

3.<https://machinelearningmastery.com/inspirational-applications-deep-learning/>

4.<https://medium.com/@vratulmittal/top-15-deep-learning-applications-that-will-rule-the-world-in-2018-and-beyond-7c6130c43b01>

ARTICLE BY

ATOM RESEARCH GROUP

PLACEMENTS

Selected Students list for



Mphasis

The Next Applied

L&T SELECTED STUDENTS

Roll.No	Name of the Student
15A81A0562	Thavanga Devi
15A81A0588	M Purna Vamsi
15A81A0517	R Satgeevathi
15A81A05G8	S Karthik
15A81A05B9	V Lasya Priya
15A81A0536	N Prudhvini Sai
15A81A05B4	S Krishna
15A81A0597	P Ravya Priya
15A81A0514	R Deepika
15A81A0546	T Neeraja
15A81A0576	R Akhila
15A81A05H0	S Sowjanga
15A81A0562	Thavanga Devi

Roll.No	Name of the Student
15A81A05F5	M.BalaKrishna
15A81A05M8	S.SaiKumar
16A81A0503	R.Ramesh
15A81A05L3	M.G.N.PavanKumar
15A81A0513	R.Bhaskar
15A81A05F1	M.Mounica
15A81A05DI	Ch.Chandini
15A81A05A2	P.Sungetha
15A81A0573	G.L.Indira
15A81A05CO	V.Indumathi
15A81A0565	Tr.Ruchitha
15A81A05A6	S.Jhansi
15A81A05D9	G.PragatiRao
15A81A0555	V.Shamitha
15A81A0526	K.Swathi
15A81A0552	V.L.Meghana
15A81A05J5	D.Rashmadhar
15A81A05KO	R.C.S.Jswarya
15A81A05M6	P.Sasanka

TCS SELECTED STUDENTS

Roll.No	Name of the Student
15A81A05Co	V.Indumathi
15A81A0546	T.Neeraja
15A81A0536	N.Prudhvini Sai
15A81A05Mo	M.H.V.Tr. Sri Vyshnavi
15A81A0501	Lakshmi Bhargavi Tr.

CAMPUS SELECTED STUDENTS



A SIX DAY ANDROID WORKSHOP BY APSSDC



GUEST LECTURE BY MR GANESH SOMISETTY SENIOR MANAGER(AIR WORLDWIDE)



EMPHASIS SELECTED STUDENTS



Answers:-

1. The boy
2. 10 fish, it is enclosed tank
3. He is dead
4. Wake up
5. No one, Robert has a sister & she is pilot
6. Out side
7. Playing chess with max
8. Foot prints
9. Cell Phone or doorbell
10. Tuesday, Thursday, today & tomorrow
11. Drop of water
12. Clock
13. The house is on the north pole; so the bear is "white"
14. Blind & he is reading Braille
15. Stop imagining

4	2	6	5	7	1	3	9	8
8	5	7	2	9	3	1	4	6
1	3	9	4	6	8	2	7	5
9	7	1	3	8	5	6	2	4
5	4	3	7	2	6	8	1	9
6	8	2	1	4	9	7	5	3
7	9	4	6	3	2	5	8	1
2	6	5	8	1	4	9	3	7
3	1	8	9	5	7	4	6	2